

WEEK 4 REPORT

Power BI : Time-Series Visualization & Financial Dashboard *(Power BI Desktop)*

Submitted by

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1. Introduction

This week I continued working on the Shopify stock dashboard from Week 3, but this time I focused on doing some exploratory data analysis (EDA) in Power BI. I calculated moving averages, built a signal to show if the stock is trending up or down, made the charts more interactive, and added a custom tooltip page. Below I have explained each step I followed.

2. Objectives

- Calculate the 20-day and 50-day Simple Moving Average (SMA) of trading volume using DAX.
- Create a Moving Average Crossover Signal to show a bullish or bearish trend (Golden/Death Cross).
- Build an interactive price trend line chart with Close Price, 20 SMA and 50 SMA, with a zoom slider.
- Build a trading volume chart with a conditional colour measure to highlight volume spikes.
- Add a date range slicer, a relative date slicer, and a parameter to switch between price metrics.
- Create a custom tooltip page that shows full trade details on hover.

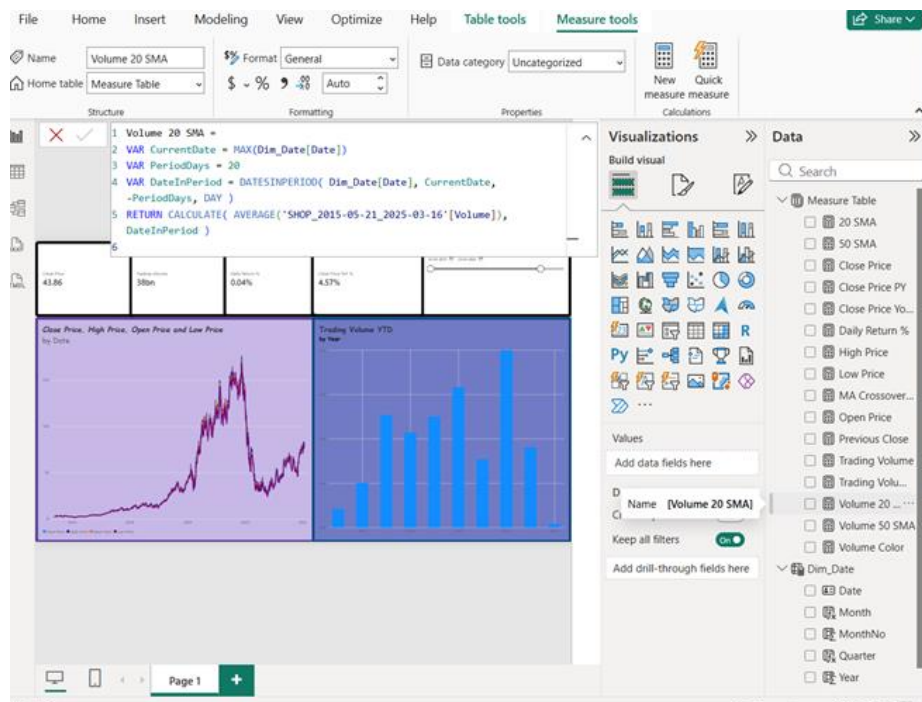
3. Steps

Step 1: Volume Moving Average (SMA) measures

I used DATESINPERIOD to calculate the average trading volume over the last 20 days and the last 50 days. This smooths out the daily ups and downs and shows the overall trend.

```
Volume 20 SMA =  
VAR CurrentDate = MAX(Dim_Date[Date])  
VAR PeriodDays = 20  
VAR DateInPeriod = DATESINPERIOD(Dim_Date[Date], CurrentDate, -PeriodDays, DAY)  
RETURN CALCULATE(AVERAGE('SHOP_2015-05-21_2025-03-16'[volume]), DateInPeriod)
```

I made the same measure for 50 days (Volume 50 SMA) by just changing PeriodDays to 50.



Step 2: Moving Average Crossover Signal

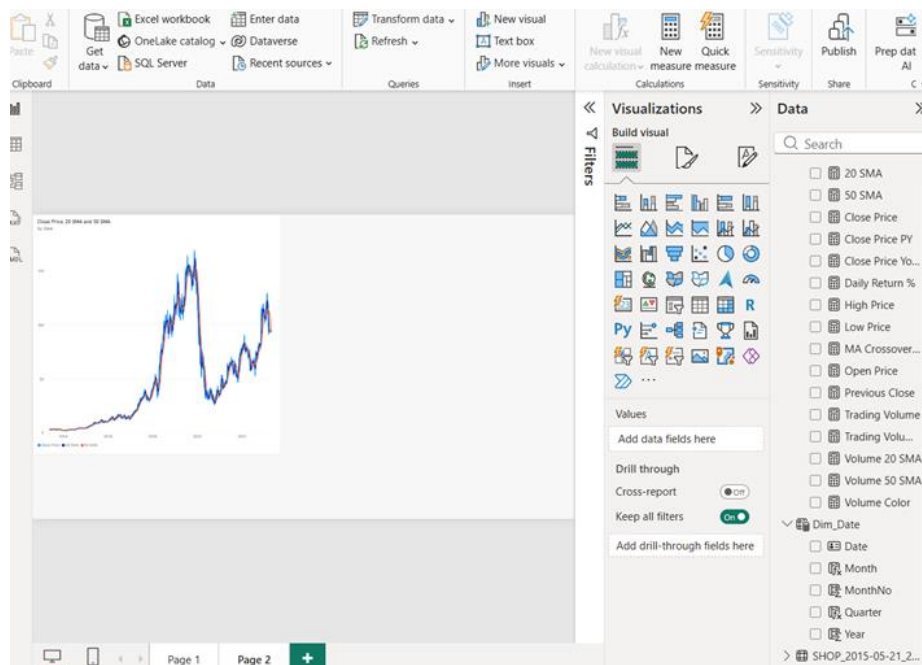
This measure compares the 20 SMA and 50 SMA and tells whether the trend is bullish or bearish. When the 20 SMA goes above the 50 SMA it is called a Golden Cross (bullish), and when it goes below it is called a Death Cross (bearish).

MA Crossover Signal =

```
IF(ISBLANK([20 SMA]) || ISBLANK([50 SMA]), BLANK(),  
  IF([20 SMA] > [50 SMA], "Bullish (20 > 50)", "Bearish (20 < 50)"))
```

Step 3: Price trend line chart

I made a line chart with Date on the X-axis (set to Continuous, so the timeline stays unbroken) and Close Price, 20 SMA and 50 SMA as the line values. I also turned on the Zoom slider under Format visual so I can quickly zoom into any time period.

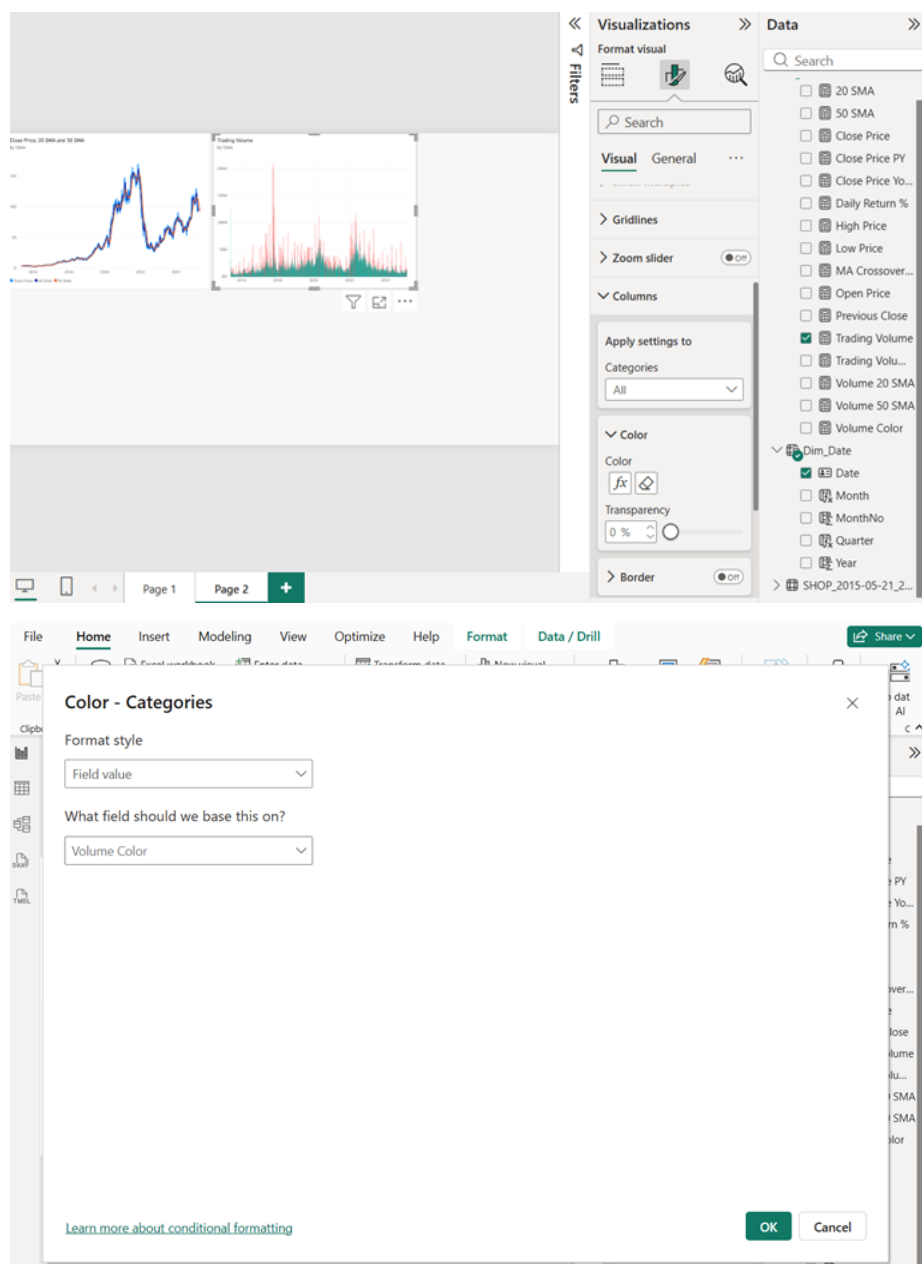


Step 4: Trading volume chart with conditional colour

Below the price chart I placed a column/area chart of Trading Volume with Volume 20 SMA, using the same continuous Date axis so both charts line up. To highlight the days with unusually high volume, I created a colour measure:

```
Volume Color = IF([Total Volume] > [20 SMA] * 1.5, "#FF4D4D", "#26A69A")
```

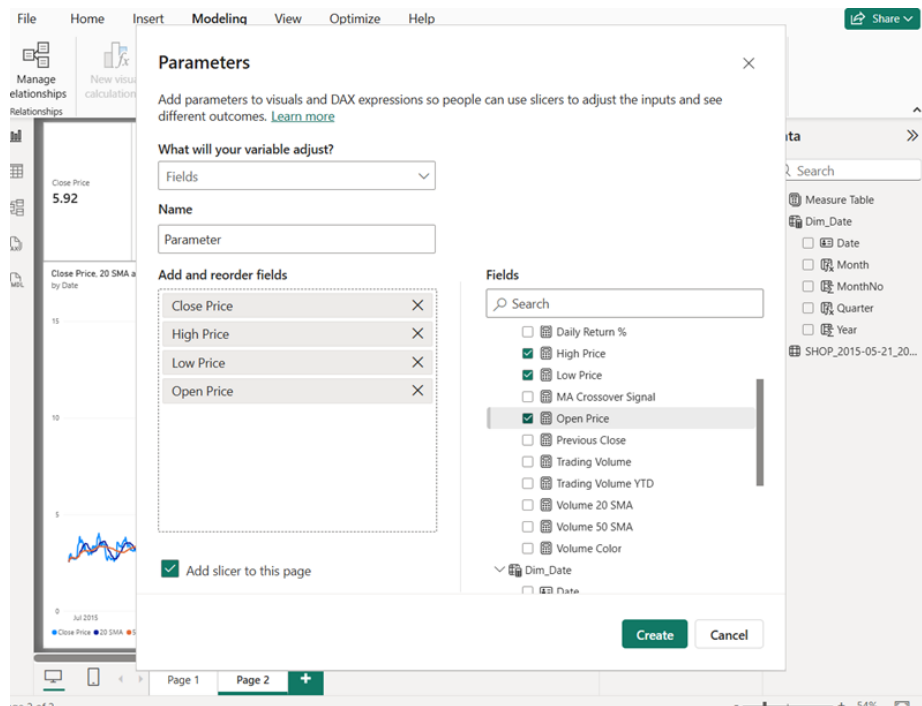
Then I applied this under Format visual > Columns > Color > Field value, and chose Volume Color as the field. This turns the volume bars red on spike days and teal on normal days.



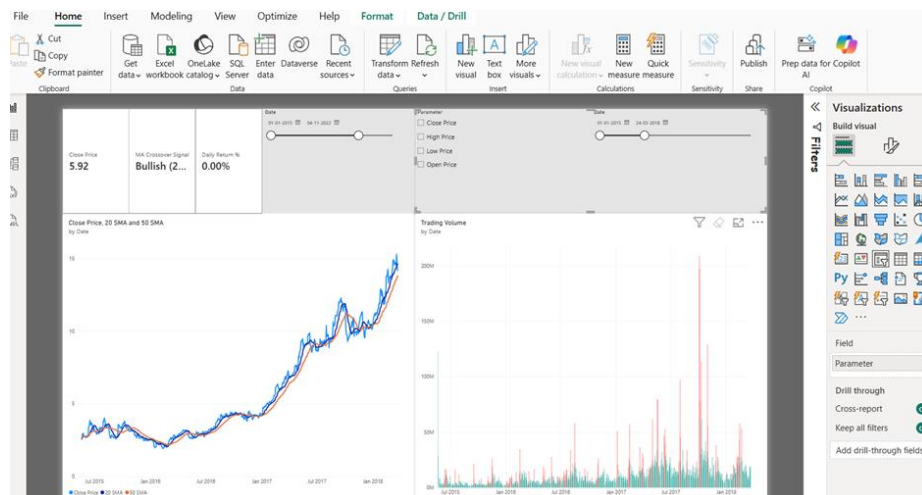
Step 5: Slicers and the metric parameter

I added a date slicer set to Between mode so I can pick any date range, and a relative date slicer for quick periods like 1 Month, 3 Months, 6 Months, YTD and 1 Year. I also

went to Modeling > New Parameter > Fields and added Close Price, High Price, Low Price and Open Price, so I can switch which price metric the chart shows using a single slicer.

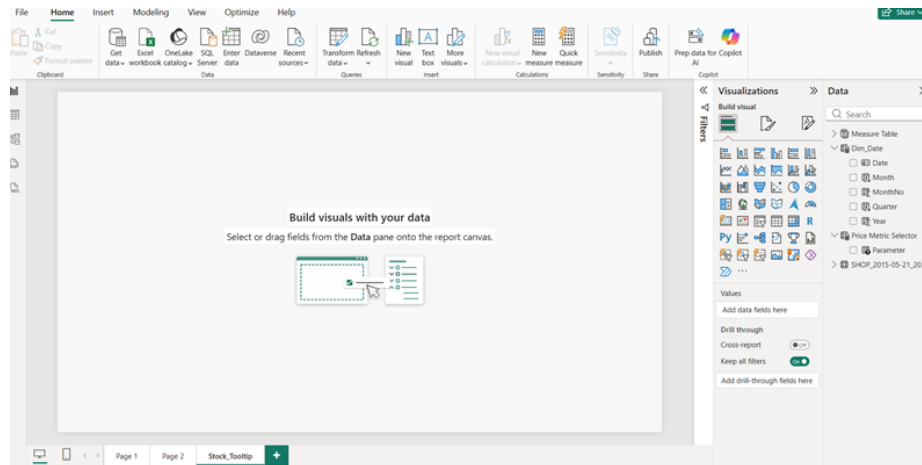


With these slicers and the parameter in place, the Stock Analysis page now looks like this:

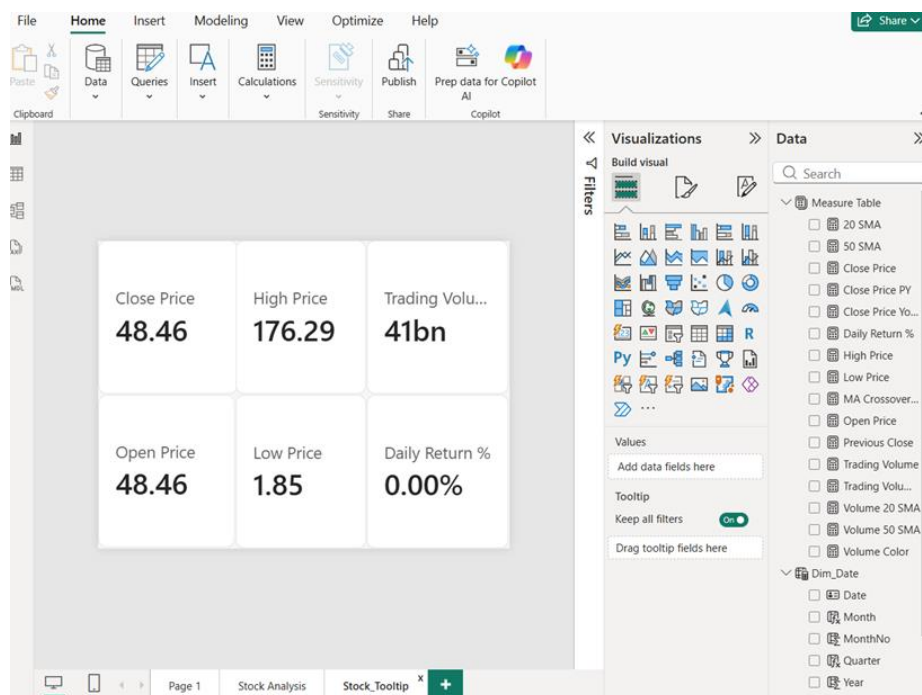


Step 6: Custom tooltip page

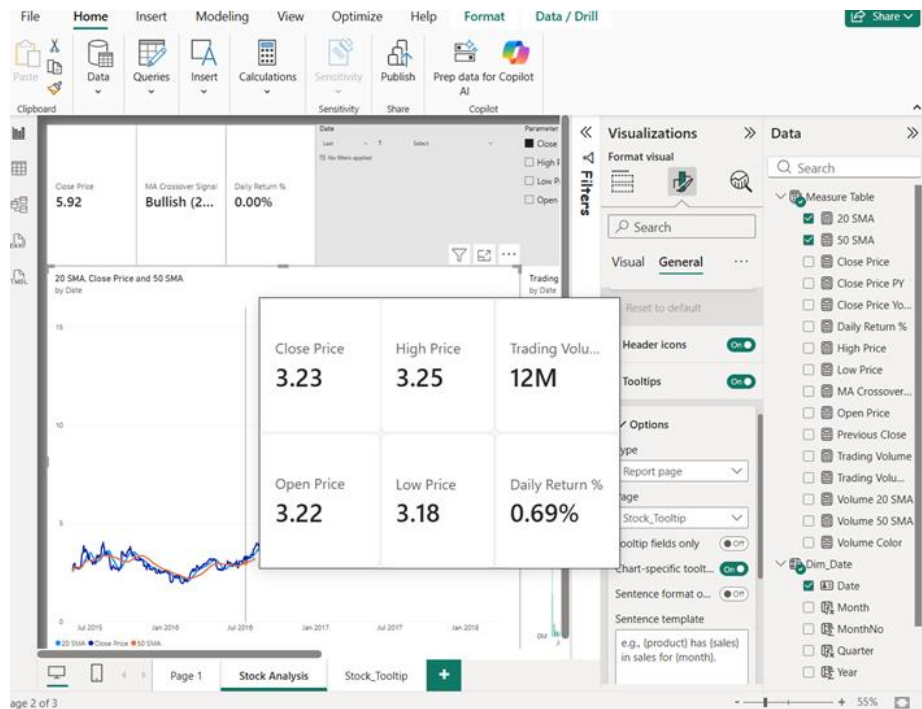
To show full trade details when hovering on the chart, I created a new page called Stock_Tooltip. In Format page > Page information I turned on "Allow use as tooltip" and set the canvas type to Tooltip.



Then I added card visuals for Close Price, High Price, Trading Volume, Open Price, Low Price and Daily Return %.

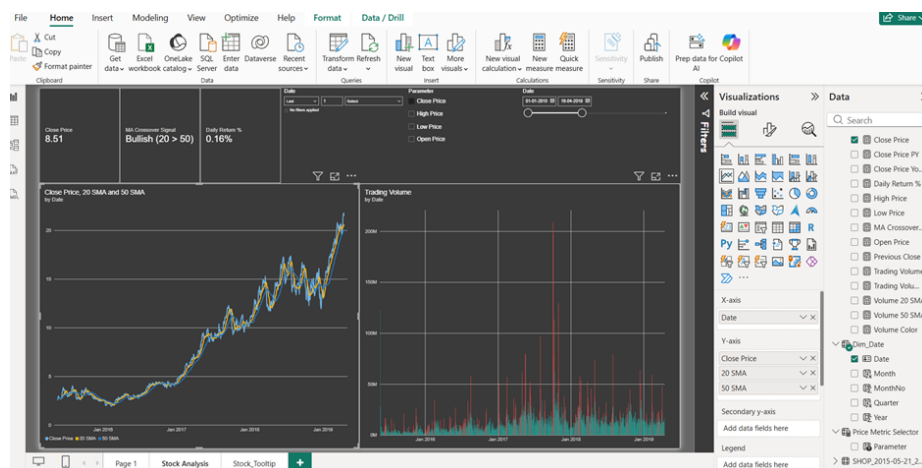


Finally, on the main price chart I went to Format visual > General > Tooltips, set Type to Page and selected Stock_Tooltip. Now hovering anywhere on the price chart shows the full detail card for that exact date.



Step 7: Final dashboard

After putting everything together, the Stock Analysis page shows the cards (Close Price, MA Crossover Signal, Daily Return %), the date and parameter slicers, the price trend line chart with the moving averages, and the coloured volume chart below it.



4. Observations

- When the 20 SMA is above the 50 SMA the signal shows Bullish, which matches the periods where the price is trending up on the chart.
- The volume chart turns red on the days with sudden spikes, and these spikes often line up with sharp moves in the price chart.

- Using the relative date slicer, it is quick to compare the last 1 month, 6 months or 1 year without changing the whole date range manually.
- The parameter switch is useful to quickly check High Price or Low Price on the same chart instead of adding separate visuals for each.
- The custom tooltip page makes it much easier to read the exact Open, High, Low, Close and Volume for a single day without adding a big table to the report.

5. Conclusion

This week I learned how to calculate moving averages with DATESINPERIOD, and how to use them to build a simple trend signal with an IF condition. I also learned how to make charts more interactive using a continuous date axis, a zoom slider, conditional colour formatting, relative date slicers and a field parameter. Building a custom tooltip page was new for me, and it made the dashboard feel much more like a real financial tool. Overall, this task helped me understand how a few DAX measures and some formatting choices can turn a plain chart into an interactive analysis dashboard.